

The digital euro has a transmission dividend, and nobody is pricing it

The digital euro is being designed around privacy, financial stability and the risk of disintermediating banks. A fourth consideration is missing from the debate: cash payments bypass the banking system through which monetary policy travels, so displacing part of cash circulation with a bank-connected instrument should make policy work better. Evidence from thirteen euro-area economies suggests the effect is real, concentrated at the horizons where policy bites hardest, and large enough that it belongs in the design discussion.

KEY MESSAGES

- Cash settles **22 per cent** of point-of-sale transactions in the Netherlands, **53 per cent** in Germany and **62 per cent** in Austria, yet the ECB sets one policy rate for all three.
- Across thirteen euro-area economies, higher cash use is associated with a **shallower output contraction** after a common monetary surprise, concentrated at quarters three to five where the average response troughs.
- The association is **not statistically significant in the baseline** and becomes so once industrial structure is controlled for. It should be read as suggestive, not settled.
- If the effect holds, the **size of the transmission gain depends on design**: holding limits and remuneration determine how much cash actually migrates into the banking system.

1 Why payment structure is a monetary variable

Monetary policy reaches the real economy through several channels, and they are not equally dependent on the medium of payment. Three of the five in the standard taxonomy operate through banks specifically: the bank-lending channel works through the supply of deposit-funded loans, the balance-sheet channel through collateral values and borrower net worth, and the deposits channel through the spread banks charge on deposit funding when the policy rate rises [1], [2]. Each presupposes that transactions and savings are routed through bank balance sheets.

A card or transfer payment satisfies that condition. A deposit is debited, a merchant account is credited, and the transaction takes place inside the financial system the central bank influences. A cash payment does not. It settles outside that system and is in that narrow sense less directly exposed to policy. Where cash is prevalent, the deposit base is smaller and less activity is bank-intermediated, so the channels depending on both should weaken.

The prediction is therefore specific rather than sweeping. Monetary policy does not stop working in cash-intensive economies; **the bank-funding-dependent channels weaken first and most**, while the interest-rate and asset-price channels are affected only indirectly. It also implies the effect should appear at the horizons where the bank-dependent channels do their work, rather than uniformly.

2 What the evidence shows

The analysis identifies monetary surprises from high-frequency changes in market rates around ECB announcements [3], estimates a structural VAR for each member state, and pools the resulting impulse responses into a single cross-country regression in which the response at each horizon is interacted with that country's cash share. Standard errors are clustered

by country. Thirteen of nineteen member states yield stable estimates on a sufficient sample, and twelve of the thirteen show output declining after a contractionary shock, which is the basic sign check that the shock is properly identified.

Constraining the cash effect to a single coefficient across all horizons produces almost nothing (0.00042, $p = 0.56$). That is itself informative: averaging across twenty-one horizons dilutes an effect theory says should be concentrated. Allowing the coefficient to vary by horizon puts it where it is predicted to be.

Horizon	Baseline	p	With manuf.	p
Quarter 3	0.00111	0.151	0.00142	0.018
Quarter 4	0.00130	0.133	0.00172	0.009
Quarter 5	0.00105	0.234	0.00153	0.023

Table 1. Effect of a one-percentage-point higher cash share on the GDP response to a one-basis-point contractionary surprise, at the trough horizons. Positive values indicate a shallower contraction. Errors clustered by country, thirteen clusters. Author's estimates.

In the baseline these coefficients are correctly signed and economically meaningful but short of conventional significance, and a joint test over the pre-specified trough window gives the same verdict (0.00115, $t = 1.50$, $p = 0.159$). Adding a control for manufacturing intensity changes that: the same three horizons return p -values of 0.018, 0.009 and 0.023.

That pattern invites an obvious objection, so it is worth addressing directly. Manufacturing intensity is a **theory-motivated control, not a result-motivated one**. Interest-sensitive durable-goods production is the standard confounder in cross-country transmission work and it correlates with cash use, so omitting it biases the cash coefficient toward zero. The trough window was likewise fixed in advance as the point where the average output response bottoms out, not selected after inspecting which horizons cooperated. And the result survives dropping any single country.

FIGURE — insert the horizon-varying cash interaction from the thesis (Figure 4, p. 23)

Figure 1. The cash interaction by horizon, with 90 per cent confidence bands. The coefficient rises into the trough and fades after quarter six, which is the shape the mechanism predicts. The shape is more informative than any single p -value.

Prices behave differently and more awkwardly. At the shortest horizon the interaction is significantly negative, a stronger price response in higher-cash countries at impact, which reads as a price-puzzle artefact rather than support for the hypothesis. The sign reverses around quarters ten to fourteen, consistent with weaker price transmission arriving late, but this is not significant under the all-items index. The honest summary is that cash appears to moderate transmission to **output at the trough**, and to prices only weakly and with a long lag.

3 What this means for the digital euro

The digital euro is a retail central bank digital currency intended to complement cash [4]. The public debate has settled on three questions: how much privacy it preserves, whether it destabilises bank funding, and how large holding limits should be. The transmission question is largely absent, and it should not be, because it runs in the opposite direction to the disintermediation worry. A digital euro that draws part of cash circulation into bank-connected form does not only risk moving deposits around; it also brings transactions that currently sit outside the transmission mechanism inside it.

Three implications follow. First, **the transmission gain is a design variable, not a fixed property**. Holding limits, remuneration and the degree of cash-likeness

determine how much cash actually migrates. A design optimised purely for privacy and cash substitution preserves the attenuation documented here; one that pulls a larger share of retail payments onto bank-connected rails reduces it.

Second, the gain is **distributed unevenly across the euro area**. On these estimates the countries standing to gain most in transmission terms are those where cash is most prevalent, which is not the same set as those most exposed to disintermediation risk. That asymmetry is politically relevant, because the cost-benefit calculation of any given design then differs systematically by member state.

Third, and most usefully for the Eurosystem, it identifies something that should be measured and currently is not. Payment structure is surveyed roughly every three years and treated as consumer-behaviour research. If it conditions the strength of transmission, it is a monetary variable and belongs in the regular monitoring set.

WHAT THIS BRIEF DOES NOT CLAIM

The estimates rest on **thirteen countries**, a small number of clusters for a cross-country interaction, and the cash share is correspondingly imprecisely estimated. The baseline output result is short of conventional significance and only clears it once manufacturing intensity is controlled for.

The design identifies an **association across countries**, not a causal effect of payment structure. Cash use is correlated with financial development, informality and the share of liquidity-constrained households, and this brief does not separate those. The price-side evidence is weaker still and should not be relied on.

Nothing here establishes **who within a country bears the attenuation**. That distributional question is the natural next step and is not answered by country-level data.

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Before this goes anywhere. Replace the figure placeholder with Figure 4 (p. 23) from the thesis. Copy the full reference entries, including volume, issue and page numbers, straight from the thesis bibliography rather than retyping them: they are already correctly formatted there, and incomplete citations are the first thing a research institute notices. Verify the Lane (2025) and ECB (2024) entries against how the thesis cites them. Then rewrite the standfirst and Section 3 in your own words, because those are the two places where the judgment has to be visibly yours.